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THE REALITY OF USING INFORMATION AND COMMUNICATION TECHNOLOGY (ICT) AND ITS RELATIONSHIP TO ACADEMIC ACHIEVEMENT AND THE DEVELOPMENT OF CREATIVE THINKING AMONG SECONDARY SCHOOL STUDENT

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ABSTRACT

Objective: This study looked into how information and communication technology (ICT) affected Jordanian secondary school pupils' academic performance and ability to think creatively.

Theoretical Framework: The purpose of the study was to evaluate how well ICT tools support students' creativity and improve their academic achievement.

Method: A balanced viewpoint on the use of ICT in education was provided by the data gathered from surveys given to 50 teachers and 50 students.

Results and Discussion: The results showed that most people had a favorable opinion of ICT's contribution to raising academic standards. Students said that using ICT tools really improved their interest in studying, motivation, and involvement. The utilization of digital technologies has been linked to better comprehension of study materials, increased growth in creative thinking abilities, and better academic achievement. It has been discovered that ICT enhances interactive and customized learning, assisting students in efficiently planning their study sessions and coming up with original solutions to challenges.

Research Implications: The study did draw attention to several obstacles, though, such as students' differing preferences for conventional teaching techniques and their challenges adjusting to new technologies. These difficulties highlighted the necessity for ongoing funding for teacher preparation programs and the creation of successful ICT integration plans. To maximize the use of technology in education, the study suggested additional investments in ICT resources and professional development for teachers.

Originality/Value: Future studies should examine the long-term effects and potency of particular ICT tools in various educational settings.

Keywords: Information and Communication Technology (ICT), academic achievement, creative thinking, education, Sustainable Development Goals (SDGs).

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1 INTRODUCTION

The role of information and communications technology is important in all aspects of life. It has helped to bring about a great civilizational shift, and there are no longer spatial or temporal barriers between members of one society, or between members of one society and another. The world has become a small electronic village; an individual can roam around it and learn about everything in it (Murad & Awda, 2014). The role of information and communication technology in the educational process has emerged since the middle of the twentieth century. It helps teachers plan and prepare their lessons, in order to present them to students in an interesting and effective way, and it also helps students teach effectively (Mensah *et al.*, 2023).

The hardware, software, network, and media components that make data collection, transmission, and processing possible are collectively referred to as information and communication technology, or ICT (Hu *et al.*, 2018). ICT is a common term used to describe the processing, manipulation, and communication of information. The use of these technologies includes the use of YouTube and other video-sharing services as well as social networking sites such as Facebook and WhatsApp. In addition to computers, other devices used for communication include projectors, desktop computers, and printers (Hu *et al.*, 2018). All of these technologies are included and used in managing information for educational purposes in the ICT mix of teaching and learning institutions. There is no doubt that the availability of information has a significant impact on students' academic achievement. More sources of information are now available to students. ICT is expanding communication options within and outside educational institutions, according to Mathivanan *et al.* (2021). This gives students—including those who are underserved by the traditional education system—access to new learning opportunities. Alderete and Formichella (2016) argue that being part of the information society means being able to access and master new technologies. This is due to the fact that ICT helps the education system to promote inclusive practices (Furlong & Davies, 2012).

This technological revolution requires providing learners with the necessary skills that develop their creative thinking, enabling them to deal with the huge amount of information, analyze, compare, synthesize and evaluate it to produce new ideas, confront life problems, and invent new methods and strategies to solve them. In response to this, the Ministry of Education in Jordan paid attention to thinking. The recommendations of the Educational Development Conference held in 1987 emphasized the necessity of developing learners' academic achievement and the development of creative thinking (Al-Nsour, 2018). Moreover, the Ministry of Education in Jordan is making great efforts to provide and employ information and communication technology in the teaching and learning process, based on introducing modern technology into education, not only in the curricula but also in all learning resource units in the school, from science labs, school libraries, resource rooms, and even school administration (Al-Omari, 2012).

1.1 PROBLEM STATEMENT AND QUESTIONS

Since its introduction, ICT has enhanced academic achievement, fostered creative thinking, and changed the way teaching and learning is conducted in advanced public schools (Clark, 2015). Others have focused on how schools view the successful integration of ICT (Ghavifekr & Rosdy, 2015). Several investigations have been conducted into how these technologies affect academic routines (Lau, Citation 2017). Research suggests that the integration of ICT in the classroom enhances students' attitudes toward academic work as well as their academic achievement (Alderete & Formichella, 2016). ICT is one of the most important factors that can help students develop their creative abilities, so ICT is now recognized as an essential resource for providing high-quality education.

Jordan is among the many countries that have made great strides in promoting the promise of ICT in creating a workforce that can fully participate in the knowledge and information society. As a result, the Jordanian government has incorporated ICT into educational policy through the Ministry of Education, ensuring that ICT is taught and studied throughout the Jordanian education system. Due to the lack of such a study in the Arab library in general, and the Jordanian library in particular, the problem of the study stems from the feeling of an urgent need to improve teaching methods and means, using modern and diverse teaching methods through the use of information and communication technology. Therefore, the research problem lies in the following main question: "What is the impact of using information and communication technology on academic achievement and the development of creative thinking among secondary school students in Jordan?"

This main question is subdivided into the following sub-questions:

1. What are the ICT tools available in Jordanian schools and what are the factors that affect the use of ICT among secondary school students?
2. What are the elements that affect ICT success in the classroom, such as student digital literacy, technological access, and instructional strategies?
3. What are the obstacles to the use of information and communication technology for teaching purposes among secondary school students from their point of view?

1.2 STUDY OBJECTIVES

The main objective of this study is: "To investigate the impact of using information and communication technology on academic achievement and the development of creative thinking among secondary school students in Jordan."

This main objective is subdivided into the following sub-objectives:

1. To reveal the effectiveness of using information and communication technology in developing some creative thinking skills among secondary school students represented in the skills of design, inference, and prediction;
2. To identify the ICT tools available in Jordanian schools and what are the factors that affect the use of ICT among secondary school students;
3. To study the elements that affect ICT success in the classroom, such as student digital literacy, technological access, and instructional strategies;

4. To identify the obstacles to the use of information and communication technology for teaching purposes among secondary school students.

1.3 STUDY SIGNIFICANCE

The importance of this research is evident by highlighting the crucial role played by ICT in teaching and learning processes at the secondary school student level. Furthermore, the results highlight the benefits of using ICT in academic achievement, developing students' creative abilities and arousing their interest in encouraging students to benefit from it. The research findings and recommendations will be useful to policy makers, educational planners, teachers and parents in developing future plans for the use of ICT in education. It helps in revealing the basic applications and software of information and communication technology that teachers use and practice for teaching purposes, in order to reveal the shortcomings in employing these applications and software, and to know their training needs to develop their uses and employ them in teaching. The research will also serve as a reference source for other studies on the subject, thus improving the quality of future research in the long run. Furthermore, the study will add to the existing knowledge, helping to bridge the gap between the benefits of ICT and its effects on academic achievement and the development of creative thinking among secondary school students.

The current research may be useful in reaching a list of creative thinking and problem-solving skills and benefiting experts in the field of education to benefit from them and put them as a focus for different curricula to develop them among students, and raise the competencies of the teaching and learning process in a manner consistent with the requirements of the era; which seeks to develop multiple patterns of thinking among the learner. This research also contributes to helping students understand modern technological phenomena and keep pace with the rapidly changing era. Finally, the results will motivate faculty members including teachers and supervisors to use ICT in both teaching and research due to its beneficial effects on the learning process.

2 LITERATURE REVIEW

2.1 ICT IN EDUCATION AND IN SCHOOLS

ICT has impacted the field of education, specifically research and teaching process (Okoro & Ekpo, 2016). According to Ashley (2016), the use of ICT in the classroom is a relevant and useful means of imparting education to students that will enhance their knowledge and equip them with the skills they need for the workforce. It provides students with a whole new and advanced learning environment; as a result, they develop a variety of skill sets in order to be productive and successful. The ability to think creatively, conduct research and evaluate information increases dramatically as students deal with larger amounts of data from a variety of sources. It is believed that the integration of ICT into the educational process makes it more complex and multidimensional as it requires teachers to have a positive mindset (Salehi & Salehi, 2012).

By providing them with better teaching materials and more effective teaching and learning strategies, ICT in schools allows teachers to adapt their methods. By providing interactive educational resources that enhance students' motivation and facilitate the acquisition of basic skills, ICT improves the learning process. More engaging and demanding learning environments are created for students of all ages through the use of various multimedia resources, including television, recordings, films and computer programs (Hussain *et al.*, 2017).

In addition, it increases the flexibility of communication education so that students can access knowledge at any time and from anywhere. Given the learner-led nature of educational procedures, which prepare students for successful learning and enhance the quality of learning, it may have an impact on the methods used in teaching and learning. The ability of students who are unable to use ICT at home to do so in the classroom is another benefit of integrating technology into the classroom. It can be used as an educational tool to deliver instruction that will improve students' learning and retention of information (Mensah *et al.*, 2023).

This technology shortens learning times while energizing and enlivening the classroom. By enhancing learner motivation and teacher training—two factors that contribute to the development of higher-order thinking abilities—ICT can improve the quality of education. ICT has fundamentally changed the way people learn, becoming an integral component of the teaching and learning process in schools (Agrahari & Singh, 2013). It is considered the most successful mass communication tool, and as such, it has significantly changed the way education is delivered. ICT is an educational tool, and Poulter and Basford (2003) argue that it has great potential to improve the standards and principles of student learning. ICT is more practical than traditional classroom instruction in terms of student achievement scores (Hussain *et al*., 2017).

2.2 THE IMPACT OF USING INFORMATION AND COMMUNICATION TECHNOLOGY (ICT) ON ACADEMIC ACHIEVEMENT AND THE DEVELOPMENT OF CREATIVE THINKING

There is accumulating data on the impact of ICT on academic achievement in poor countries (Avgerou, 2010; Walsham, 2017). A student's overall ICT proficiency is essential to their use of technology. An empirical study by Karamti (2016), which examined the impact of ICT on academic achievement in Tunisia, revealed that Tunisian parents primarily considered ICT to be important for their children's academic success, despite substantial ethical concerns about students' greater use of technology. Al-Ansi *et al*. (2019) also examined the impact of ICT on different learning environments in developing countries. They argued that the use of ICT has brought traditional teaching techniques into the modern era of interactive environments. They again discovered how ICT is the engine that drives the revolution in education. ICT has a positive and significant impact on learning in secondary school. ICT elements had a positive and significant impact on the learning process at both undergraduate and postgraduate levels, with the exception of devices and gadgets, which were negative at undergraduate level and negligible at both levels (Mensah *et al*., 2023).

This shows that academic achievement and ICT use are important topics being researched in emerging countries. Furthermore, Nigerian families believe that their children's ability to acquire ICT knowledge depends on how successful they will be in school and the future workforce (Emeka & Nyeche, 2016). These findings suggest that despite some skepticism, families embrace and encourage the increasing use of ICT for educational purposes. According to Rakhyoot (2017), who investigated the institutional and individual challenges of adopting e-learning in higher education in Oman, academics' views, e-learning has been successful in facilitating student communication, teamwork and positive learning attitudes. In addition, research by Darkwa and Antwi (2021) titled "From Classroom to Online: Comparing the Effectiveness and Academic Performance of Students in Classroom and Online Learning in Ghana" indicated a positive relationship between online learning and improved academic performance (Al-Nsour, 2018).

The impact of children's daily interactions with technology has gained prominence as a topic of study in the context of academic learning, indicating the inherent learning potential of ICT use. Furlong and Davies (2012) use in-depth qualitative analysis in a significant mixed method review that provides compelling evidence of the benefits of ICT use settings in the UK. Traditional categories of formal and informal learning, such as schoolwork and homework, have been replaced by semi-formal and incidental learning, such as studying while having fun and relaxing, according to Furlong and Davies' hypothesis. This tends to suggest that children are learning continuously through their daily experiences with technology rather than through focused and planned work in school as they integrate digital information into their daily lives.

In order to provide strong evidence for the benefits of ICT-enabled environments in the UK, Furlong and Davies (2012) conducted a comprehensive mixed method review that used in-depth qualitative analysis. Furlong and Davies' idea is that traditional categories of formal and informal learning, such as schoolwork and homework, have been replaced by semi-formal and incidental learning, such as studying while having fun and relaxing. As digital information is integrated into everyday life, this means that children learn through technology on an ongoing basis through daily interactions rather than through focused and temporary schoolwork. According to a study by Skryabin *et al*. (2015), who used data from 120 countries, including developing countries such as Ghavifekr and Rosdy (2015), continuous digital learning enhances students' growth in reading, mathematics and science. Furlong and Davies (2012) describe this phenomenon as a blurring of the boundaries between home and school. They go on to say that this is because most people acquire ICT literacy skills at home. This illustrates how the use of ICT can enhance a student's ability to learn in general, even when it comes to extracurricular activities such as social networking and surfing the Internet (Lee & Wu, 2013).

Larry (2003) believes that the teacher can contribute to the development of creative thinking among students by giving them enough time to think, and providing reinforcement and rewards for ideas. He also creates a positive environment in the classroom by having a quiet class that is characterized by acceptance and non-coercion, and providing rich and effective stimuli and excluding fear and failure. Harris (2004) identified some trends that students must embody in order to be creative, which are: curiosity, challenge, perseverance and flexible imagination. Educators place their hopes on information and communication technology in developing creative thinking skills among students, as developing thinking and problem-solving skills is one of the most important goals of computerizing education in Jordan (Al-Omari, 2012).

Slipchuk *et al*. (2020) describe innovative teaching strategies in universities, such as the use of ICT. Dotsenko *et al*. (2023) describes the technology used to produce educational content for free digital resources in general technical disciplines. Fatima and Jabeen, (2021) consider the relationship between the academic achievement of university students and their use of ICT (Zahorodnia *et al*. 2024).

Rubab (2021) evaluates the impact of ICT use on the achievement of university students. The use of ICT in preparing students for research work is examined in the research paper (Information and Communication Technologies in Preparing Students for Research Work, 2019). Hasanova (2022) studies how students perceive information in relation to ICT. Tashtoush *et al*. (2023) investigate the impact of ICT learning on academic interest in mathematics.

Gurevich *et al*. (2021) examine the application of augmented reality technology in higher education. Pydiura (2020) details how folklore is studied in Ukrainian educational institutions using digital technologies and virtual reality. Al Farsi *et al*., (2021) provide an overview of the application of virtual reality in education. Volynets (2021) provides an overview of how virtual reality technologies are used in education. Al-Muqaddam (2019) covers the application of augmented and virtual reality in education. Yassin and Hisham (2021) provide an overview of the literature on the application of augmented and virtual reality in the classroom.

3 METHOD

In order to find out how information and communication technology (ICT) affects academic performance and the growth of creative thinking in Jordanian secondary school pupils, this study used a descriptive analytical technique. There were 100 participants in the study, 50 of whom were secondary school teachers and the other 50 were students. To guarantee a varied representation of the educational situation, the participants were chosen from a variety of secondary schools located throughout Jordan. Instructors were chosen based on their background using ICT in the classroom, and students were picked to represent a variety of academic backgrounds and levels of expertise using digital learning resources. Statistical techniques, such as mean scores and standard deviations, were employed to examine quantitative data in order to evaluate the overall effect of ICT on students' academic performance and creative thinking.

4 RESULTS AND DISCUSSION

4.1 DESCRIPTIVE OF DEMOGRAPHIC VARIABLES

Table 1

Category characteristics

	Frequency	Percent	Valid Percent	Cumulative Percent
Teacher	50	50.0	50.0	50.0
Students	50	50.0	50.0	100.0
Total	100	100.0	100.0	

As can be seen from Table (1), teachers and students made up an equal portion of the respondents, accounting for 50% of the sample as a whole. 50 instructors and 50 students made up the total number of participants (100), ensuring a balanced distribution between the two groups.

Table 2

Gender characteristics of teachers

	Frequency	Percent	Valid Percent	Cumulative Percent
Male	33	66.0	66.0	66.0
Female	17	34.0	34.0	100.0
Total	50	100.0	100.0	

The gender distribution of teachers is seen in Table (2). 34% of the 50 teachers are female, and 66% of the teachers are male. This demonstrates that little over one-third of the sample's teachers are female, with men making up over two-thirds of the total.

Table 3

Level of experience for the teachers

	Frequency	Percent	Valid Percent	Cumulative Percent
Less than 1 year	7	14.0	14.0	14.0
1-5	13	26.0	26.0	40.0
6-10	18	36.0	36.0	76.0
More than 10 years	12	24.0	24.0	100.0
Total	50	100.0	100.0	

The experience levels of the teachers are shown in the table. Of the fifty teachers, 14% have worked for less than a year, 26% have worked for one to five years, 36% have worked for six to ten years, and 24% have worked for ten years or more.

Table 4

Gender characteristics of students

	Frequency	Percent	Valid Percent	Cumulative Percent
Male	28	56.0	56.0	56.0
Female	22	44.0	44.0	100.0
Total	50	100.0	100.0	

The student's gender distribution is shown in Table (4). Of the 50 pupils, 46% are female and 56% are male. This indicates that there are somewhat more male pupils in the group than female students.

Table 5

Grade of students

	Frequency	Percent	Valid Percent	Cumulative Percent
First Secondary	16	32.0	32.0	32.0
Second Secondary	34	68.0	68.0	100.0
Total	50	100.0	100.0	

Table (5) shows the distribution of students by their grade level. Of the 50 students, 32% are in the first secondary grade, while 68% are in the second secondary grade. This indicates that the majority of students are in the second secondary grade.

4.2 INVESTIGATING THE IMPACT OF USING INFORMATION AND COMMUNICATION TECHNOLOGY ON ACADEMIC ACHIEVEMENT AND THE DEVELOPMENT OF CREATIVE THINKING AMONG SECONDARY SCHOOL STUDENTS IN JORDAN

Through the use of SPSS, the descriptive statistics (means and standard deviation) of the responses and their ranks, which were obtained using a five-point Likert scale, were calculated. Means between (1 and 1.80) were regarded as very low, between (1.81 and 2.60) as low, between (2.61-3.40) as a medium, between (3.41-4.20) as high, and between (4.21-5.00) as very high.

Table 6

Investigating the impact of using information and communication technology on academic achievement and the development of creative thinking among secondary school students in Jordan

No.	Statements	Means	Standard deviations	Practices degree
Investigating the impact of using information and communication technology on academic achievement and the development of creative thinking among secondary school students in Jordan				
Q1	There is a significant increase in students' desire to use information and communication technology to learn online	3.58	0.727	High
Q2	There is a possibility to increase the desire and motivation for education using modern technologies in education	3.71	0.729	High
Q3	The use of information and communication technology helps to stimulate the learner's motivation and needs, and form his new trends.	3.71	0.729	High
Q4	The use of ICT helps to gain experiences and increase the learner's active participation.	3.77	0.694	High
Q5	The use of ICT helps to engage the learner's senses, which leads to consolidating understanding and comprehension.	3.66	0.699	High
Q6	The use of ICT helps to confront the problem of verbal errors and form sound concepts.	3.54	0.915	High
Q7	The use of ICT helps to confront individual differences among learners and modify their behavior.	3.64	0.927	High
Q8	The use of ICT helps to organize and continue the ideas formed by the learner.	3.72	0.740	High
Q9	The use of ICT helps to make the educational material more understandable and more resistant to forgetting.	3.73	0.709	High
Q10	The use of ICT has helped in individual learning	3.78	0.660	High
Q11	The use of ICT increases the thinking strategy	3.71	0.743	High

Q12	The use of ICT increases the mental effectiveness of the individual learner	3.70	0.785	High
Q13	The use of ICT increases the motivation and enhances the learner's self-satisfaction	3.77	0.664	High
Q14	The use of ICT helps to help the learner retain what he has learned	3.72	0.740	High
Q15	The use of ICT helps save time and effort for the learner.	3.77	0.790	High
Q16	The use of ICT helps develop the learners' ability by increasing their ability to distinguish and classify their sensory perceptions.	3.72	0.766	High
Q17	ICT works as a method for solving problems that arise for learners.	3.84	0.707	High
Q18	ICT allows learners to have the ability to remember information longer.	3.79	0.591	High
Q19	ICT allows learners to have a greater ability to organize educational material and motivates learners and attracts them to the lesson.	3.85	0.702	High
Q20	ICT develops learners' positive tendencies and motivates them to learn by doing.	3.66	0.742	High
Q21	ICT contributes to increasing the strength of learners' personality and developing their creative thinking.	3.72	0.740	High
Q22	ICT contributes to developing various cognitive, linguistic, behavioral, educational and social aspects of learners.	3.78	0.690	High
Q23	The use of ICT helps in understanding the study materials better	3.76	0.726	High
Q24	The use of ICT contributes to improving the grades.	3.82	0.657	High
Q25	The use of ICT enhances the ability to think creatively	3.91	0.637	High
Q26	The use of ICT contributes to the development of creative skills in a positive way.	3.91	0.621	High
Q27	The use of ICT provides greater opportunities to develop new ideas and creative solutions	3.81	0.647	High
Q28	The use of ICT helps in organizing study time	3.87	0.677	High
Q29	The use of ICT helps develop creative skills by providing new tools and resources.	3.79	0.701	High
	Overall	3.75	0.122	High

According to the survey, secondary school pupils in Jordan have a generally positive perception of the contribution that information and communication technology (ICT) makes to improving their academic standing and encouraging original thought. The findings demonstrate a deep appreciation for the ways in which ICT enhances educational opportunities, inspires pupils, and fosters innovation. Overall, the results indicate that students recognize the importance of digital technologies in the classroom and view ICT as a vital tool for improving their cognitive and academic development.

Even while some students still have difficulties or prefer traditional teaching methods, one important result of ICT is that it greatly boosts students' enthusiasm to learn using online platforms. This indicates that students are becoming more accustomed to and open to using digital resources, which emphasizes how crucial it is to assist them in adjusting to new teaching strategies. Although some students might face challenges, the encouraging trend suggests that a large number of students are embracing online platforms as an integral part of their educational journey. The results also show that ICT increases student involvement and improves the enjoyment of learning. Schools may build dynamic and engaging learning environments that grab students' attention and encourage a positive attitude toward education by utilizing contemporary educational technologies. The findings imply that incorporating ICT into instructional strategies can enhance student engagement and motivation by making learning more enticing.

Furthermore, the study demonstrates that ICT has a big impact on how pupils develop their long-term learning habits. Beyond instant interaction, ICT technologies facilitate the formation of fruitful study habits by connecting course content to relevant, real-world applications. Long-term student interest maintenance is crucial to establishing academic achievement over the long term, and this link between learning and real-world application can support it.

Additionally, students admit that using ICT enables them to actively engage in their education as opposed to just passively receiving knowledge. Digital tools provide students with dynamic and useful apps that motivate them to interact more thoroughly with the material. Given that active engagement frequently results in improved retention and comprehension of the content, this is especially crucial for promoting deeper understanding and improving the learning process as a whole. The combination of interactive, aural, and visual components is one of the multimodal aspects of ICT that seems to help pupils grasp difficult ideas. The study emphasizes how crucial sensory involvement is to learning and how digital technologies can support a variety of learning preferences. This is essential for enhancing students' understanding and memory retention as well as providing more meaningful reinforcement for previously learned material.

Finally, the study highlights how ICT fosters original thought and problem-solving skills. Pupils understand the importance of digital technologies in fostering creativity, exploring novel approaches, and honing critical thinking abilities. One of the main advantages of integrating ICT into the curriculum is that it fosters the creative and cognitive development that students need to thrive in a world that is becoming more and more reliant on technology, in addition to helping them achieve academic success.

The present study's results are consistent with previous research in highlighting the beneficial effects of ICT on students' academic achievement. In the study, secondary school students in Jordan demonstrated a clear awareness of the function that ICT plays in advancing their academic performance and bettering their comprehension of study materials. In a similar vein, research highlights how ICT enhances learning outcomes for students by offering superior instructional resources and encouraging more in-depth interaction with the material. According to Mensah *et al*. (2023), ICT helps pupils become more motivated and acquire necessary skills, which improves academic achievement. The study and earlier research emphasize how much better pupils can understand complex subjects when they use digital tools.

The contribution of ICT to encouraging students' creative thinking is another obvious commonality. The current study emphasizes the role that ICT plays in fostering creativity and emphasizes how crucial it is for assisting students in coming up with original ideas and solutions. This result is consistent with the research, which highlights how ICT provides interactive resources and exploration opportunities that allow students to participate in creative processes. Al-Omari (2012) states that ICT is seen as an essential tool in Jordan for promoting creative thinking and problem-solving abilities, which lends more credence to the current study's conclusions that technology inspires pupils to think creatively.

The benefits of ICT for motivation in the learning process are highlighted in both the current study and the literature review. Students in the survey said that using ICT in the classroom increases their motivation and level of participation. This is consistent with the literature, where researchers like Salehi and Salehi (2012) highlight how ICT makes learning settings more dynamic and interesting and inspires students to participate more actively in their education. According to Hussain *et al.* (2017), the research also reveals that ICT can boost students' motivation by providing them with multimedia tools that grab their interest and make studying more pleasurable. This alignment shows how widely held the belief is regarding the contribution of ICT to more accessible and interesting learning.

The study's conclusions and the literature diverge in a few ways despite these commonalities. According to the current study, ICT can assist bridge the achievement gaps between individuals. Pupils saw ICT as a crucial instrument for customized learning, enabling them to study in ways that best fit their requirements and at their own speed. Though the research acknowledges the flexibility that ICT offers, it focuses more on the larger institutional obstacles associated with integrating technology in the classroom, including issues related to teacher preparation and infrastructure constraints (Tezci, 2011). The benefits of ICT are acknowledged in both the study and the literature; however, the literature often focuses on systemic hurdles to its adoption, which were not a major emphasis of the current study's findings.

5 CONCLUSION AND RECOMMENDATIONS

The purpose of this study was to investigate how information and communication technology (ICT) affects Jordanian secondary school students' academic performance and capacity for creative thought. ICT significantly improves kids' academic performance and creativity, according to the research. Students experienced increased motivation, engagement, and a more dynamic learning environment when different technological tools were incorporated into the teaching and learning process. Using ICT tools promoted creative thinking and problem-solving abilities in addition to helping students comprehend and retain study content better. The overall favorable opinion of ICT's function in education is highlighted by the comprehensive mean score of 3.75 across many ICT-related criteria.

Surveys of instructors and students were used as part of the approach to find out how they felt about the use of ICT in the classroom. The results showed that ICT tools, like online platforms and multimedia materials, are essential for raising students' motivation to learn, enhancing their academic achievement, and promoting the growth of their capacity for creative thought. Notably, the study found that ICT promotes customized learning, accommodates various learning styles, and allows interactive learning experiences—all of which help the development of both cognitive and creative abilities.

Although the study yielded favorable results, it also revealed certain obstacles, like students' differing inclinations towards conventional vs digital teaching approaches and certain hurdles in adjusting to novel technologies. Optimizing the use of ICT in education requires addressing these issues. According to the research, even while ICT improves learning and creativity, in order to fully realize ICT's potential in the classroom, there is a need for ongoing investments in teacher preparation programs and the creation of efficient teaching techniques.

Based on the results the study recommended that:

1. Continuous professional development is necessary to help teachers successfully integrate ICT tools into their lesson plans;
2. Schools should create a welcoming atmosphere that promotes the use of digital tools while resolving any reluctance or issues that students might have with it;
3. It is also advised that ICT techniques and tools be regularly reviewed and updated to make sure they still match students' changing demands and improve their educational experiences.

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